

Maintaining the Jack Mackerel Technical Annex

A handover for the next assessment scientist

Jack Mackerel Working Group

2026-09-15

1 Start here

The maintained annex is `technical-annex-v2/` in the **SC14_JM** repository. Open `output/technical-annex.html`, `.pdf` or `.docx`. Edit the plain-text files in `report/` and rebuild from this folder:

```
Rscript build.R          # annex and this guide in HTML
Rscript build.R all       # HTML, PDF and Word
Rscript build.R pdf annex # one document and format
Rscript build.R check     # verify inputs and risk calculations
```

The build uses R and Quarto. PDF also needs XeLaTeX and the fonts in `report/_quarto.yml`. Installation commands are in `README.md`. The supplied `software/jjmR_1.2020.1.tar.gz` preserves the local package fixes used by the annex. The build records package versions, logs and product hashes in `validation/`.

These are the saved September 2026 Model 1.06 working results. This migration changes the workflow and location; assessment findings and decision status remain as recorded in the annex. Running the build reads saved fits, calculates summaries and draws figures. A new assessment or projection requires a separate JJM run.

2 Follow the calculation

Source	R calculation	Result
<code>data/assessment/config/</code> , <code>input/</code> , <code>results/</code> <code>Base msy_mt block</code> <code>data/projections/*_R.rep</code>	<code>R/prepare_report.R</code> reads four fits with <code>jjmR::readJJM</code> Mean of column 10 for 2017–2026 <code>R/risk_data.R</code> reads projected SSB and ADMB standard error	Model objects, status and catch summaries in <code>data/derived/</code> Fixed BMSY for stock status and risk All 180 annual records in <code>data/derived/risk-</code> <code>data.csv</code>
Saved fits and retrospectives	R chunks in <code>report/annex_tables.qmd</code> and <code>annex_plots.qmd</code>	Report tables and figures
Text, tables and figures	Quarto assembles the includes in <code>report/technical-</code> <code>annex.qmd</code>	HTML, PDF and Word in <code>output/</code>

2.1 Example probability

For each stock, scenario and year, `R/risk_data.R` calculates:

```
100 * pnorm(BMSY, mean = projected_SSB, sd = ADMB_SE,  
            lower.tail = FALSE)
```

The fixed target uses the **base** assessment; the projected biomass and standard error use the **.ls** ten-year projection. All quantities are in kt. The result is an annual conditional normal probability. The calculation holds BMSY fixed and excludes uncertainty in that threshold. R recalculates every record and compares it with the reviewed export before the report uses it.

For single-stock scenario 4 in 2028, the target is 6,869.029 kt, projected SSB is 5,774.3 kt and its standard error is 1,147.25 kt. The result is 16.999%, displayed as **17%**. The tables show 2028, 2031 and 2036: two, five and ten years after 2026.

2.2 Keep the reference points distinct

The fixed risk target is a **2017–2026 mean of annual base BMSY**. The model's separate **global scalar BMSY** uses 2011–2025 mean selectivity. Scenario 4 optimizes and applies its projection FMSY with 2023–2025 mean future selectivity. The equations and scalar values are in `report/annex_reference_points.qmd`; `data/projections/manifest.json` identifies the saved projection run and template.

The original five-year projection plots remain historical comparisons. The risk tables use the separate ten-year products. Three explanatory diagrams and the reference-value tables retain reviewed September values; inspect their labels when changing models. Original diagram drawing scripts are preserved in the archive. These saved elements require an explicit update when new runs are adopted.

3 Make a change

For prose, edit `report/annex_text.qmd` or the relevant included section. For literature, edit `report/refs.bib`. Keep figure and table labels stable. After a Word review, transfer agreed edits back into the `.qmd` source and rebuild.

For a new assessment, complete and review the model run first. The optional R helper copies this release's explicit set of 24 assessment files into a **new** folder and checks that the fits and saved objects can be read:

```
Rscript R/snapshot_assessment.R /path/to/jjm /path/to/new-snapshot
```

Review that candidate before replacing `data/assessment/`. A new model or year requires reviewing the helper's selection, year/model settings, recruitment periods, retrospective years, reference values and projection horizons. Update the ten-year run products and their uncertainty calculation separately. Record the source revision and changes, then update `data/input-checksums.csv` only for the reviewed inputs. The inventory is deliberately checked rather than rewritten by a routine build.

4 Check and hand over

Run `Rscript build.R check`, rebuild the required formats, and inspect status, reference and risk tables, figure captions and links. Build logs are in `validation/logs/`. `validation/build.csv` records each product's date and hash; `validation/R-session.txt` records the build environment. The migration check records what was compared with the previous package. Rendering and automated checks provide only part of accessibility validation; reading order and assistive technology still require review.

Keep the whole `technical-annex-v2/` folder together. Its data links are relative, and the build needs no other local checkout. Historical sources and concise reviews are isolated in `archive/september-12-history.zip`. The parent repository's wiki has a separate build; it is independent of the annex's R workflow. Commit reviewed source and generated reports together when ready, and treat publication as a separate step.